

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B5. GENERAL RELATIVITY

TRINITY TERM 2023

Monday, 12 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

Expressed in terms of the Christoffel symbols $\Gamma_{\beta\gamma}^\alpha$, the Riemann tensor has components

$$R^\gamma_{\alpha\beta\delta} = \frac{\partial}{\partial x^\delta} \Gamma_{\alpha\beta}^\gamma - \frac{\partial}{\partial x^\beta} \Gamma_{\alpha\delta}^\gamma + \Gamma_{\alpha\beta}^\epsilon \Gamma_{\delta\epsilon}^\gamma - \Gamma_{\alpha\delta}^\epsilon \Gamma_{\beta\epsilon}^\gamma.$$

The Ricci tensor is defined as $R_{\alpha\beta} \equiv -R^\gamma_{\alpha\beta\gamma}$. When expressed in a basis in which the metric tensor $g_{\alpha\beta}$ is diagonal, it can be written as

$$R_{\alpha\beta} = \frac{1}{2} \frac{\partial^2}{\partial x^\alpha \partial x^\beta} \ln |g| - \frac{\partial \Gamma_{\alpha\beta}^\gamma}{\partial x^\gamma} + \Gamma_{\alpha\gamma}^\delta \Gamma_{\beta\delta}^\gamma - \frac{1}{2} \Gamma_{\alpha\beta}^\gamma \frac{\partial}{\partial x^\gamma} \ln |g|,$$

where $|g|$ is the modulus of the determinant of $g_{\alpha\beta}$.

The Einstein equation is

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R - \Lambda g_{\alpha\beta} = -\frac{8\pi G}{c^4} T_{\alpha\beta},$$

where $R \equiv g^{\alpha\beta} R_{\alpha\beta}$ is the Ricci scalar, Λ is the cosmological constant and $T_{\alpha\beta}$ is the energy-momentum tensor.

For the line element

$$ds^2 = -c^2 dt^2 + R^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

the Einstein equation can be written as the pair of equations

$$\begin{aligned} \ddot{R} &= -\frac{4\pi G}{3} \left(\rho + \frac{3p}{c^2} \right) R + \frac{1}{3} \Lambda c^2 R, \\ \dot{R}^2 &= \frac{8\pi G}{3} \rho R^2 + \frac{1}{3} \Lambda c^2 R^2 - c^2 k, \end{aligned}$$

where ρ and p are the rest-frame density and pressure, respectively.

1. Einstein's theory of gravity is given by the field equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = -\kappa_E T_{\mu\nu}$$

where $\kappa_E = 8\pi G/c^4$.

(a) Use the weak field, static limit, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and the weak field limit of the Ricci tensor given by

$$R_{\mu\nu} = \frac{1}{2}\eta^{\alpha\gamma}(-\partial_\mu\partial_\alpha h_{\nu\gamma} + \partial_\gamma\partial_\alpha h_{\nu\mu} + \partial_\mu\partial_\nu h_{\alpha\gamma} - \partial_\nu\partial_\gamma h_{\alpha\mu})$$

to show that, in the presence of a non-relativistic source such that $T_{00} = \rho c^2$ and $T_{ij} = 0$, we recover the Newton-Poisson equation, $\nabla^2\Phi = 4\pi G\rho$, where $h_{00} = -2\Phi/c^2$. [6]

(b) Show that, in the weak field limit, and keeping terms up to first order in u^i (where $u^\alpha = dx^\alpha/dt$) the equation of motion of a particle is given by

$$\frac{d^2\vec{x}}{dt^2} = -\vec{\nabla}\Phi + \vec{v} \times \vec{B}_G,$$

where $\vec{v} = d\vec{x}/dt$, $\vec{B}_G = \vec{\nabla} \times \vec{h}$ and $\vec{h} \equiv (h_{01}, h_{02}, h_{03})$. [Hint: use the vector identity $[\vec{A} \times (\vec{C} \times \vec{B})]_i = A_j B_i C_j - A_j B_j C_i$] [7]

(c) The angular momentum of the Earth is $J = 2MR^2\omega/5$ where R , M and ω are the radius, mass and angular velocity of the Earth. At a distance r from the centre of the Earth we have that $\vec{B}_G$ is a dipole field with an amplitude given by $2GJ/c^2 r^3$. Approximately compare the acceleration due to $\vec{B}_G$ with that of a satellite in geosynchronous orbit. [3]

(d) Consider now an alternative theory of gravity proposed by Gunnar Nördstrom with field equations given by

$$R = -\kappa_N g_{\mu\nu} T^{\mu\nu}$$

and where the metric takes the simple form $g_{\mu\nu} = e^{2\Phi/c^2}\eta_{\mu\nu}$ in which Φ is a function of space-time and $\eta_{\mu\nu}$ is the flat space metric, and particles in free fall follow geodesics. Use the weak field limit (i.e. the expression for $R_{\mu\nu}$ given in part (a)) to show that, in the non-relativistic limit we recover the Newton-Poisson equation and find κ_N in terms of κ_E . [5]

(e) Explain why there is no deflection of light and why $\vec{B}_G = 0$ in this theory. [4]

2. The geodesic deviation equation for two adjacent geodesics, $x^\alpha(\lambda)$ and $x^\alpha(\lambda) + \Delta x(\lambda)$ is given by

$$\frac{D^2 \Delta x^\lambda}{d\tau^2} - R^\lambda_{\alpha\nu\beta} \dot{x}^\alpha \dot{x}^\beta \Delta x^\nu = 0.$$

(a) Adopting a Local Inertial Frame in the first instance, use the geodesic equation to derive the geodesic deviation equation, as given above. Explain on what basis you can claim that your result applies in an arbitrary frame. Explain also how this equation connects gravitation and space-time. [8]

(b) Consider a space-time with a metric of the form

$$ds^2 = -(1 - F)c^2 dt^2 + \frac{dr^2}{1 - F} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

where $F = F(r)$. Show that for a massive particle in an equatorial orbit, the following equations are satisfied

$$(1 - F)\dot{t} = k, \quad r^2 \dot{\phi} = h,$$

where $\dot{X} = dX/d\tau$, τ is proper time and h and k are constants. [4]

(c) For the case $F(r) = r_s/r$ where r_s is a constant, show that a massive particle obeys

$$\dot{r}^2 - \frac{r_s c^2}{r} + \frac{h^2}{r^2} - \frac{r_s h^2}{r^3} - c^2(k^2 - 1) = 0.$$

Define an effective potential for the orbits in this space. Find the radius of the stable and unstable orbits in this potential and explain what the last stable orbit is. Find the radius of the last stable orbit and the angular momentum of a particle that is on it. Show that, for this orbit, $k = 2\sqrt{2}/3$. [6]

(d) A massive particle falls radially into the black hole. Assume the particle has no energy as $r \rightarrow \infty$. Find the particle trajectory in terms of proper time τ . What happens to the coordinate velocity, dr/dt , of the particle as it reaches $r = r_s$? [7]

3. Consider a vacuum space-time (i.e. where $T_{\mu\nu} = 0$) with the following metric

$$ds^2 = -c^2 dt^2 + \alpha^2 \left(e^{2\beta} dx^2 + e^{-2\beta} dy^2 \right) + dz^2$$

where α and β are functions of $ct + z$ only.

- (a) Consider a coordinate transformation to a new coordinate system (u, v, x, y) , where $u = ct + z$ and $v = ct - z$. Show that the metric becomes

$$ds^2 = -dudv + \alpha^2 \left(e^{2\beta} dx^2 + e^{-2\beta} dy^2 \right).$$

Calculate the Christoffel symbols corresponding to the above metric.

[7]

- (b) Show that a curve having $\dot{t} = -1$ and $\dot{x} = \dot{y} = \dot{z} = 0$ is a timelike geodesic (i.e. geodesic and $g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} < 0$).

[3]

- (c) Show that the only non-zero component of the Ricci tensor is given by

$$R_{uu} = -2 \left(\frac{\alpha''}{\alpha} + \beta'^2 \right), \quad (1)$$

where the prime indicates d/du . Explain how, for a given $\beta(u)$ and initial conditions for α , we can solve the Einstein field equations to find $\alpha(u)$.

[7]

- (d) Now take $\beta(u) = 0$ and $\alpha(u) = 1$ for $u < 0$ and $\beta(u) = u$ for $u \geq 0$. Find $\alpha(u > 0)$, and hence the metric, and describe what happens to a ring of particles in the (x, y) plane.

[5]

- (e) Describe what properties of this metric allow us to interpret it as an exact, transverse, gravitational wave and how it differs from a gravitational wave in linearized general relativity.

[3]

4. Imagine a positively-curved Universe with a metric

$$ds^2 = -dt^2 + R^2 \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

and which contains both non-relativistic matter and dark energy with an energy density that evolves as $\rho_E \propto R^{-n}$.

(a) Use conservation of energy and momentum to find the equation of state for the dark energy. [4]

(b) Starting with the Friedmann equation, show that

$$\left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G \rho_0}{3R^3} \left[1 - \frac{R}{R_T} + \left(\frac{R}{R_E} \right)^{3-n} \right],$$

where ρ_0 , R_T and R_E are constants. Derive the corresponding equation for $\ddot{R}/R$. [4]

(c) Consider the case $n = 0$. Let R_S be the special value of R for which the Universe doesn't evolve (i.e. it neither expands nor contracts). Find R_S and the corresponding value of R_E as a function of R_T . This is Einstein's Static Universe. Write down the Friedmann equation as a mechanical system of the form

$$E = \frac{1}{2} \dot{R}^2 + V_{\text{eff}}(R) = \frac{8\pi G \rho_0}{3R_T}.$$

Use the effective potential, $V_{\text{eff}}(R)$ to show that Einstein's Static Universe is unstable. Describe how the Universe will evolve if $\dot{R} = 0$ and R is not exactly R_S . [5]

(d) Consider now the case $n = 2 - \epsilon$ where $0 < \epsilon \ll 1$. As for part (c), find the special value of R , for which such a Universe doesn't evolve. Use the effective potential for this model to describe the evolution of such a Universe when it has enough kinetic energy to just overshoot R_S . This is known as a Loitering Universe. [6]

e) Taylor expand the effective potential to 2nd order around R_S so that it looks like an inverted harmonic oscillator. Use this approximate form of the Friedmann equation to show that the time it takes for the universe to expand from R_S to $2R_S$ tends to infinity as $\epsilon \rightarrow 0$. [6]